

Airline Delay Analysis: Predicting Flight Delay Propagation Through Data-Driven Analysis

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Abstract—Flight delays continue to create operational and scheduling challenges for airlines, airports, and passengers. This project develops a machine learning approach for predicting whether a flight will be delayed by 20 minutes or more using historical airline performance data. The dataset was preprocessed by removing invalid records, handling missing values, selecting relevant operational features, and converting categorical variables through one-hot encoding. A Decision Tree Classifier using Gini impurity was implemented and evaluated using accuracy, precision, recall, F1-score, and confusion matrix metrics. Initial results demonstrate that interpretable classification models can identify useful patterns associated with flight delays while also providing feature importance rankings for operational analysis.

Index Terms—flight delay prediction, decision tree, gini impurity, classification, machine learning, airline performance

I. INTRODUCTION

In the aviation industry, flight delays and cancellations are a persistent issue which often results in passengers being required to adjust plans without warning. These disruptions are influenced by a variety of factors including operational changes, weather fluctuations, and scheduling conflicts. While the causes of these delays are diverse, patterns and correlations can be found within the flight delay data, indicating the likelihood of certain delays occurring.

This project aims to analyze delays and identify varying indicators of impending flight delays, then draw correlations between flights delays. The referenced data set includes information on expected and actual departure and arrival times, various types of flight delays, and the times at which flight delays occurred. By examining these relationships, the project aims to better understand how delays are connected and how they may propagate within airline systems.

To support this analysis, the flight data will be preprocessed and cleaned to ensure consistency. Relevant operational and scheduling variables are selected, while categorical variables such as airline carrier and airport codes are transformed using one-hot encoding. A Decision Tree Classifier using Gini impurity is then trained to predict whether a flight will arrive at

least 20 minutes late. The resulting model also provides feature importance rankings, allowing identification of the strongest contributors to flight delays.

II. LITERATURE REVIEW

Recent research on flight delay analysis indicated that delays are influenced by a combination of factors, primarily consisting of operational changes, weather fluctuations, and scheduling conflicts. Machine learning analysis and prediction methods have been widely applied to model and predict the causes of these flight delays. The following studies were selected based on their analysis of similar data sets or their application of similar machine learning methodology.

Jadhav *et al.* proposed a multitier ensemble model, A-MERIT-C, for real-time flight delay prediction using the Airline On-Time Performance dataset [1]. Their approach incorporates one-hot encoding, streaming data techniques, and ensemble learning to classify flight delays. The model achieved approximately 80.15% accuracy with relatively low evaluation time (under 13 seconds). This study is directly relevant because it demonstrates the effectiveness of the same dataset used in this project and highlights the importance of pre-processing. Additionally, the emphasis on real-time data reveals that delay patterns can change over time, supporting the need to analyze relationships between delay types rather than relying solely on static predictive models.

AlBassam and AlShahrani examined the impact of class imbalance on flight delay prediction by applying resampling techniques such as Random Oversampling, SMOTE, and ADASYN [2]. The study evaluated multiple machine learning models, including Decision Trees, Random Forests, Support Vector Classifiers, Logistic Regression, K-Nearest Neighbors, and Naive Bayes. The results showed that Random Forest with Random Oversampling and Decision Tree with SMOTE achieved the highest accuracy, approaching 90%. This work highlights the importance of data pre-processing, particularly in handling imbalanced datasets. The findings are relevant

to this project because uneven distributions of delay types can affect correlation analysis and conditional probability calculations, making proper data preparation essential.

Phisannupawong *et al.* investigated flight delay prediction using advanced modeling techniques that integrate multiple data sources and representations [3]. Their approach focuses on improving predictive performance through more complex feature extraction and learning methods. While the techniques used are more advanced than those implemented in this project, the study emphasizes the importance of incorporating diverse and meaningful features when analyzing flight delays. This supports the inclusion of time, operation, and delay related variables in the current analysis to emphasize relationships between delay types.

Yigma analyzed airline on-time performance and its impact on passenger behavior and operational efficiency [4]. The study identified key factors influencing delays, including weather conditions, airport congestion, scheduling constraints, and prior delays. In addition, it found that delays are more frequent during peak travel periods and certain seasons. Most importantly, the study demonstrated that delays can propagate throughout the system, where one delay leads to subsequent delays. This concept of delay propagation is central to the current project, as it directly aligns with the goal of analyzing how different delay types influence one another.

Overall, these studies indicate that flight delays are not isolated events; they are interconnected through multiple influencing factors. They highlight important challenges such as class imbalance, evolving data patterns, and the need for effective pre-processing of data. These findings support the use of correlation analysis and conditional probability in this project to better understand relationships between delay types, while also justifying the use of machine learning methods as supporting tools for identifying patterns and validating results.

III. IMPLEMENTATION

The implementation of this project focuses on building a decision tree which can predict if flights will be delayed significantly (delays of 20 minutes or more) using historical airline data from the past 20 years. The full implementation is completed in python due to its extensive machine learning and visualization libraries. Several libraries were used in development; Pandas, NumPy, Scikit-learn, and Matplotlib were the primary libraries used. NumPy was used for numerical operations and to process and handle missing values. Scikit-learn was used for the machine learning components, including the Decision Tree Classifier, prediction generation, and evaluation metrics such as accuracy, precision, recall, and F1-score. Matplotlib was used to generate visual outputs for experimentation and progress reporting. These tools together provided the foundation for preprocessing, model development, experimentation, and visualization.

The dataset used in the project was stored in the file `airline.csv.shuffle`, which holds the various previous flight records. Each row in the dataset represents one flight and includes scheduling information, actual delay and arrival times,

various delay measures, airport identifiers, carrier identifiers, aircraft identifiers, and a variety of calendar data. Due to the size of the dataset (12 GB), the overall dataset is randomly sampled for 100,000 rows, which are stored in the `airline.csv.shuffle` and directly loaded into memory. After loading the file into memory, the data is cleaned by removing duplicate rows, nonsensical numerical data (such as negative delays), and flights which were cancelled or diverted. In addition, data columns which gave away the answer to whether or not a flight was delayed, such as the total delay, were also removed. After preprocessing and feature selection, 19 core features were retained, including temporal, operational, and categorical variables such as departure time, airline carrier, origin, destination, and aircraft identifier. Numeric data columns in entries were converted from strings to integers. Missing numerical values were replaced by the median imputation and missing categorical values were assigned the most common category. Finally, categorical features were transformed using one-hot encoding. All of this pre-processing allows for repeat testing while preserving the diversity and quality of the data and avoiding model overfitting.

After the file is initially loaded into memory, the dataset is split into 3 sets (training, testing, and validation) with another round of random sampling. The training set contains 70% of the data, while the testing and validation sets are both 15%. The training set is used to find, through machine learning, the patterns which result in significant flight delays. The validation set is used to determine model performance after initial learning and to fine-tune decisions. Finally, the testing set is used to show model functionality and performance on the data. The 3-way split helps reduce overfitting and enforce reliable performance measures.

The target variable for classification was created using the arrival delay field. Flights arriving less than 20 minutes late were assigned class 0, representing on-time or minor delay performance. These are not considered significant. However, flights arriving 20 minutes late or more were assigned class 1, representing a meaningful operational delay. This converted the problem into a binary classification task suitable for supervised machine learning.

The machine learning model chosen for implementation was a Decision Tree Classifier using the Gini impurity criteri:

$$Gini = 1 - \sum_{i=1}^n p_i^2$$

where p_i represents the proportion of samples belonging to class i at a given node. Decision trees were selected because they are interpretable, efficient, and able to model nonlinear decision boundaries without requiring feature scaling. The implemented tree used a maximum depth of 10, minimum split size of 20 samples, minimum leaf size of 10 samples, and a fixed random state for reproducibility. During training, the model recursively selected feature splits that most reduced impurity, creating decision rules that separate delayed and non-delayed flights. Initial experimentation has been completed successfully. The model was trained on the processed training

set and evaluated on the validation set using classification metrics. Accuracy measures the overall percentage of correct predictions. Precision measures how often predicted delays were actually delays. Recall measures how many real delayed flights were successfully detected. F1-score balances precision and recall into a single measure. A confusion matrix was also generated to summarize correct and incorrect classifications across both classes. These metrics provide an early indication of model effectiveness and areas for future tuning.

Because the model was implemented using scikit-learn, feature importance values were extracted directly from the trained classifier using the `feature_importance_` attribute. The analysis showed that departure-related variables, particularly `DepDelay` and scheduled departure time (`CRSDepTime`), were among the most influential predictors. Additionally, encoded categorical variables such as airline carrier and airport identifiers contributed significantly to prediction performance, supporting the hypothesis that both operational conditions and airline-specific factors influence delay outcomes.

Figure 1 displays the number of flights in each target class after preprocessing. This visualization helps identify whether the dataset is balanced or if one class appears significantly more often than the other. Detecting class imbalance is important because it can affect training behavior and metric interpretation.

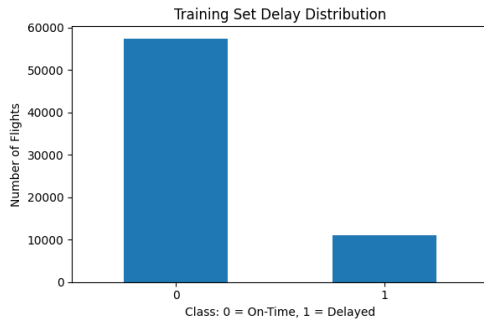


Fig. 1. Training Set Delay Distribution

Figure 2 summarizes the model's validation performance using a bar chart. This provides a clear comparison of the classifier's strengths and weaknesses during initial experimentation.

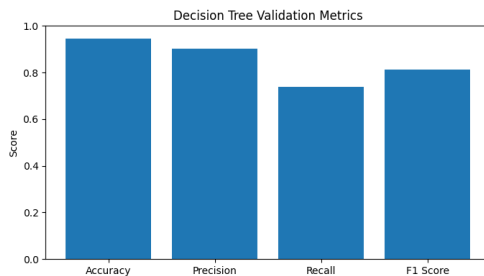


Fig. 2. Decision Tree Validation Metrics

Figure 3 displays the confusion matrix as a heatmap showing true positives, true negatives, false positives, and false negatives. This helps explain what kinds of prediction mistakes the model makes most often.

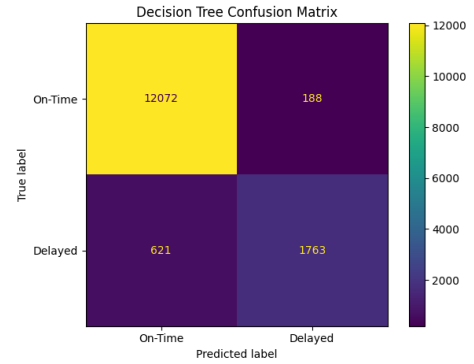


Fig. 3. Decision Tree Confusion Matrix

Overall, the implementation has progressed beyond planning and into successful experimentation. The system now supports data loading, preprocessing, feature engineering, binary classification, validation testing, prediction generation, and model interpretation. Future work will focus on tuning tree parameters, comparing alternative models such as Random Forests, and expanding experimentation using larger portions of the original dataset.

IV. CONCLUSION

This project demonstrates that a machine learning pipeline implemented in Python using scikit-learn can effectively predict significant airline delays based on historical flight data. By sampling a manageable subset of the dataset, applying data cleaning techniques, and transforming categorical variables using one-hot encoding, a structured feature matrix was constructed for model training.

A Decision Tree Classifier using the Gini impurity criterion was successfully trained and evaluated using a three-way data split. The model achieved meaningful performance across accuracy, precision, recall, and F1-score metrics, while also providing interpretable decision rules. Feature importance analysis revealed that departure-related variables, airline carriers, and airport identifiers play a significant role in predicting delays.

Although the current implementation uses a sample of the full dataset, the results indicate that decision tree models are capable of capturing key delay patterns. Future work will focus on scaling the model to larger datasets, improving handling of class imbalance, and comparing performance with more advanced ensemble models such as Random Forests and Gradient Boosting. Overall, this study highlights the effectiveness of data-driven approaches for understanding and predicting flight delay behavior.

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